

## Electronic Circuits

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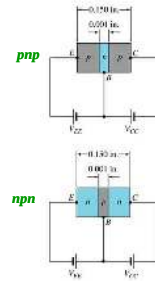
## Transistor Construction

There are two types of transistors:

- *pn*p
- *np*n

The terminals are labeled:

- E - **E**mitter
- B - **B**ase
- C - **C**ollector

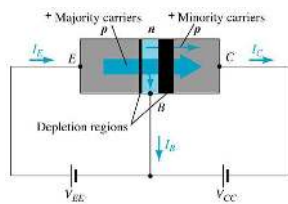


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## Transistor Operation

With the external sources,  $V_{EE}$  and  $V_{CC}$ , connected as shown:

- The emitter-base junction is forward biased
- The base-collector junction is reverse biased



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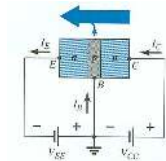
## Currents in a Transistor

Emitter current is the sum of the collector and base currents:

$$I_E = I_C + I_B$$

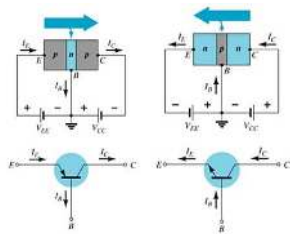
The collector current is comprised of two currents:

$$I_C = I_{C_{\text{majority}}} + I_{C_{\text{minority}}}$$



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## Common-Base Configuration



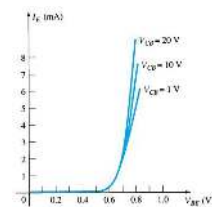
The base is common to both input (emitter-base) and output (collector-base) of the transistor.

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## Common-Base Amplifier

### Input Characteristics

This curve shows the relationship between of input current ( $I_E$ ) to input voltage ( $V_{BE}$ ) for three output voltage ( $V_{CB}$ ) levels.

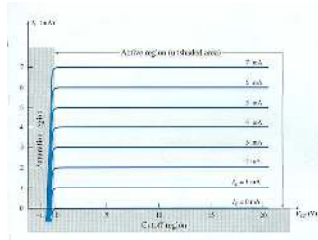


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## Common-Base Amplifier

### Output Characteristics

This graph demonstrates the output current ( $I_C$ ) to an output voltage ( $V_{CB}$ ) for various levels of input current ( $I_E$ ).



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## Operating Regions

- **Active** – Operating range of the amplifier.
- **Cutoff** – The amplifier is basically off. There is voltage, but little current.
- **Saturation** – The amplifier is full on. There is current, but little voltage.

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## Approximations

Emitter and collector currents:

$$I_C \approx I_E$$

Base-emitter voltage:

$$V_{BE} = 0.7 \text{ V (for Silicon)}$$

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## Alpha ( $\alpha$ )

Alpha ( $\alpha$ ) is the ratio of  $I_C$  to  $I_E$  :

$$\alpha_{dc} = \frac{I_C}{I_E}$$

Ideally:  $\alpha = 1$

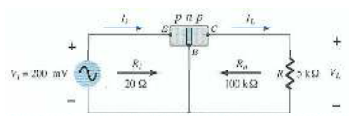
In reality:  $\alpha$  is between 0.9 and 0.998

Alpha ( $\alpha$ ) in the AC mode:

$$\alpha_{ac} = \frac{\Delta I_C}{\Delta I_E}$$

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## Transistor Amplification



Currents and Voltages:

$$I_E = I_i = \frac{V_i}{R_i} = \frac{200 \text{ mV}}{20 \text{ k}\Omega} = 10 \text{ mA}$$

$$I_C \approx I_E$$

$$I_L \approx I_i = 10 \text{ mA}$$

$$V_L = I_L R = (10 \text{ mA})(5 \text{ k}\Omega) = 50 \text{ V}$$

Voltage Gain:

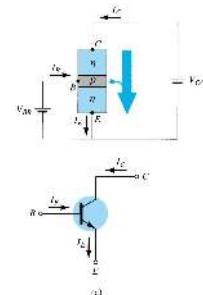
$$A_v = \frac{V_L}{V_i} = \frac{50 \text{ V}}{200 \text{ mV}} = 250$$

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## Common-Emitter Configuration

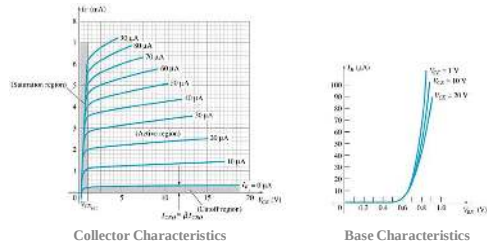
The emitter is common to both input (base-emitter) and output (collector-emitter).

The input is on the base and the output is on the collector.



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## Common-Emitter Characteristics



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## Common-Emitter Amplifier Currents

### Ideal Currents

$$I_E = I_C + I_B$$

$$I_C = \alpha I_E$$

### Actual Currents

$$I_C = \alpha I_E + I_{CBO} \quad \text{where } I_{CBO} = \text{minority collector current}$$

$I_{CBO}$  is usually so small that it can be ignored, except in high power transistors and in high temperature environments.

When  $I_B = 0 \mu A$  the transistor is in cutoff, but there is some minority current flowing called  $I_{CEO}$ .

$$I_{CEO} = \frac{I_{CBO}}{1 - \alpha} \quad I_B = 0 \mu A$$

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## Beta ( $\beta$ )

$\beta$  represents the amplification factor of a transistor. ( $\beta$  is sometimes referred to as  $h_{fe}$ , a term used in transistor modeling calculations)

In DC mode:

$$\beta_{dc} = \frac{I_C}{I_B}$$

In AC mode:

$$\beta_{ac} = \frac{\Delta I_C}{\Delta I_B} \bigg|_{V_{CE} = \text{constant}}$$

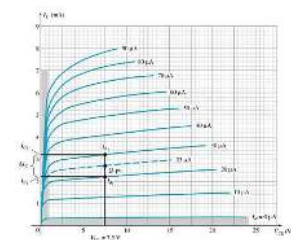
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## Beta ( $\beta$ )

Determining  $\beta$  from a Graph

$$\begin{aligned} \beta_{AC} &= \frac{(3.2 \text{ mA} - 2.2 \text{ mA})}{(30 \mu A - 20 \mu A)} \\ &= \frac{1 \text{ mA}}{10 \mu A} \bigg|_{V_{CE} = 7.5} \\ &= 100 \end{aligned}$$

$$\begin{aligned} \beta_{DC} &= \frac{2.7 \text{ mA}}{25 \mu A} \bigg|_{V_{CE} = 7.5} \\ &= 108 \end{aligned}$$



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## Beta ( $\beta$ )

Relationship between amplification factors  $\beta$  and  $\alpha$

$$\alpha = \frac{\beta}{\beta + 1} \quad \beta = \frac{\alpha}{\alpha - 1}$$

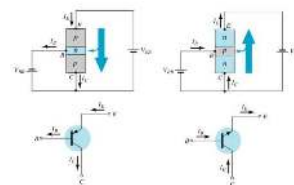
Relationship Between Currents

$$I_C = \beta I_B \quad I_E = (\beta + 1) I_B$$

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## Common-Collector Configuration

The input is on the base and the output is on the emitter.



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# Transistor Terminal Identification

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